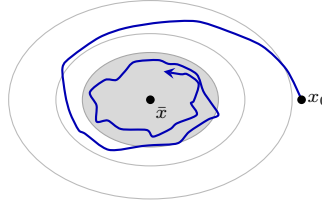


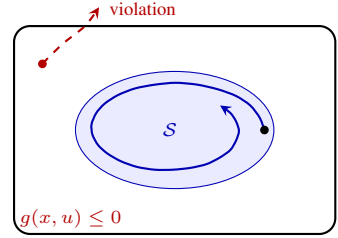
$$\Delta V(x_k) \leq -\alpha_3(\|x_k - \bar{x}\|)$$

(a) Nominal stability



$$\text{radius } \gamma(\sup_j \|w_j\|) \text{ or } \gamma(\text{tr } \Sigma)$$

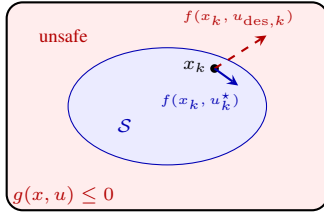
(b) Robust and stochastic stability



$$g(x, u) \leq 0$$

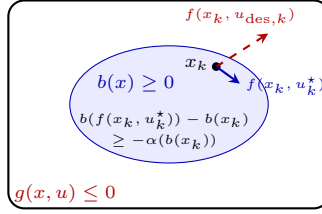
forward invariance

(c) Invariance and safety



maximal safe set

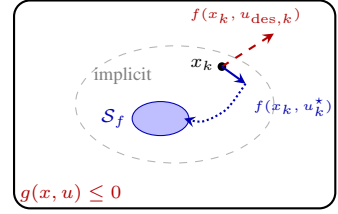
(d) Hamilton-Jacobi reachability



$$g(x, u) \leq 0$$

one-step condition

(e) Control barrier function



$$g(x, u) \leq 0$$

backup trajectory

(f) Predictive safety filter

Fig. 1: Illustration of the stability and safety notions of this section. (a) Nominal stability: a Lyapunov function  $V$  decreases along closed-loop trajectories (level sets in gray) and the state converges to the equilibrium  $\bar{x}$ . (b) Robust and stochastic stability: under disturbances  $w$ , the state converges to a neighbourhood of  $\bar{x}$  (shaded), whose size depends on the disturbance bound or the noise covariance. (c) Safety: constraint satisfaction at one time instant does not imply safety (red dashed); safety is guaranteed by an invariant set  $\mathcal{S}$  contained in the constraints. (d)–(f) Safety filters: the desired input  $u_{\text{des},k}$  would leave the safe set and is modified to  $u_k^*$ . HJ reachability computes the maximal safe set  $\mathcal{S}$  offline and its complement (red) contains the states from which a constraint violation cannot be avoided. A CBF certifies invariance of  $\{b(x) \geq 0\}$  by the one-step condition on  $b(x_{k+1})$ . A PSF does not compute the safe set (dashed) but certifies safety by a backup trajectory (dotted) into a smaller terminal set  $\mathcal{S}_f$ .